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Fine-Grained Image-Text Correspondence with Cost Aggregation for Open-Vocabulary Part Segmentation

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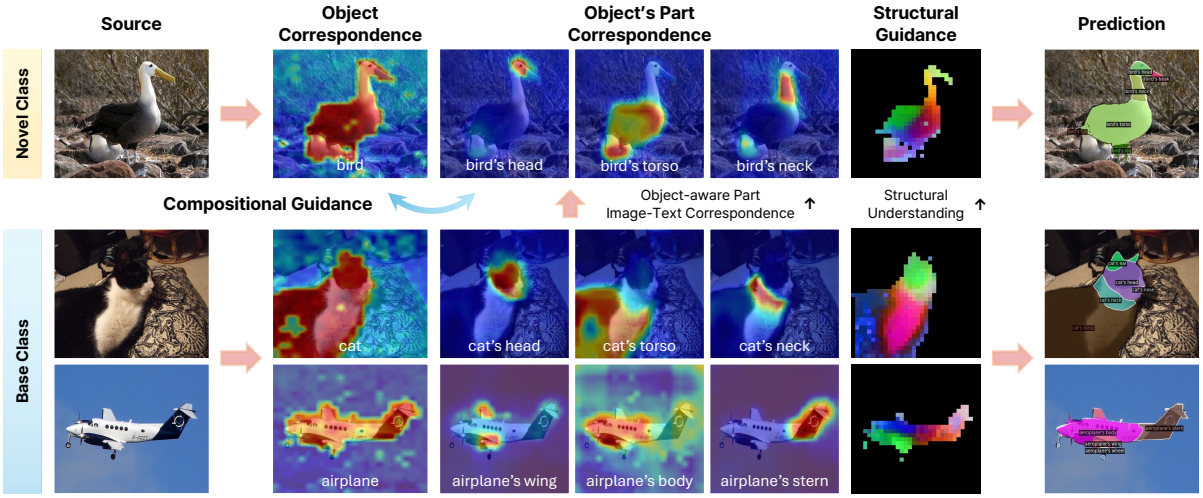


Figure 1. The proposed PartCATSeg exploits part-level and object-level image-text correspondence using cost aggregation, enhancing object-aware part image-text matching. Additionally, it utilizes structural guidance to achieve successful part segmentation even for classes not encountered during training.

Abstract

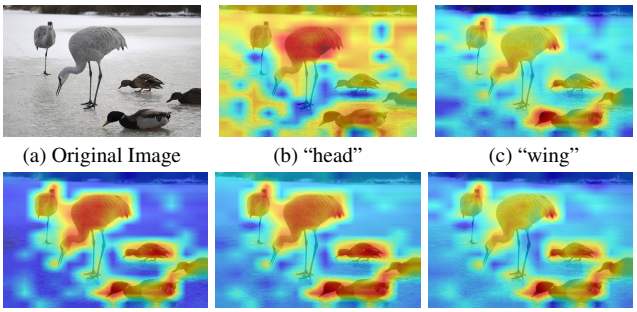
Open-Vocabulary Part Segmentation (OVPS) is an emerging field for recognizing fine-grained parts in unseen categories. We identify two primary challenges in OVPS: (1) the difficulty in aligning part-level image-text correspondence, and (2) the lack of structural understanding in segmenting object parts. To address these issues, we propose PartCATSeg, a novel framework that integrates object-aware part-level cost aggregation, compositional loss, and structural guidance from DINO. Our approach employs a disentangled cost aggregation strategy that handles object and part-level costs separately, enhancing the precision of part-level segmentation. We also introduce a compositional loss to better capture part-object relationships, compensating for the limited part annotations. Additionally, structural guidance from DINO features improves boundary delineation and inter-part understanding. Extensive experiments on Pascal-Part-116, ADE20K-Part-234, and PartImageNet datasets demonstrate that our method significantly outperforms state-of-the-art approaches, setting a new baseline for robust generalization to unseen part categories.

1. Introduction

The rapid advancements in model architectures and improved training techniques have allowed cutting-edge models to achieve exceptional performance on closed-set datasets and object-level segmentation [6, 9, 10, 26, 34, 46, 55]. However, they still exhibit suboptimal performance in zero-shot scenarios [1, 2, 4, 20] and open-vocabulary setups [22, 38, 69, 74, 76]. Moreover, part-level fine-grained segmentation [3, 8, 24] remains challenging, highlighting the ongoing difficulties in these research areas.

Recently, Open-Vocabulary Part Segmentation (OVPS) has emerged as a novel approach aimed at recognizing novel objects at a fine-grained, part level [14, 36, 57, 65]. OVPS enables the segmentation of detailed object parts, even for

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(a) Original Image (b) “head” (c) “wing”

(d) “bird” (e) “bird’s head” (f) “bird’s wing”

Figure 2. **CLIP Image-Text Similarity Visualization for Object-Level and Part-Level Text.** The visualization [37] compares the frozen CLIP [53] image-text similarity between object-level and part-level text descriptions. (a) shows the original images; (b), (c) depict the part-level similarities for terms such as ”head” and ”wing” while (e), (f) show object-specific parts. The stronger activation for object-level text in (d) suggests a dominant focus on the entire object rather than individual parts in the image-text correspondence.

categories not seen during training, positioning it as both a research challenge and a high-demand application area. This framework has numerous practical applications, including robotic control [63, 70], medical imaging [71], image editing [40, 44], and image generation [32, 64], serving as a significant milestone for the advancement of visual systems.

OVPS research has recently accelerated, leveraging pretrained vision-language models (VLMs) [31, 35, 53] to transfer part-level knowledge from base to novel classes. Recently, VLPart [57] utilizes DINO [5, 50] features to establish part-level correspondence between base and novel categories, while OV-PARTS [65] focuses on leveraging object-level contextual information to enhance part-level segmentation. More recent approaches, such as PartGLEE [36] and PartCLIPSeg [14], further advance the field by employing attention-based approaches with joint training of object and part to refine part segmentation.

Despite recent advancements, OVPS still suffers from poor performance, as key challenges remain unresolved. First, aligning part-level text with its corresponding visual features is more challenging than at the object level. This stems from the significantly lower ratio of part-level text-image pairs compared to object-level text during the pretraining of VLMs, as noted in previous studies [24, 51, 57]. Consequently, part-level data often receives weaker or noisier guidance compared to object-level data due to its diversity. Additionally, the limited availability of part-level ground truth (GT) labels in base datasets further degrades performance. Figure 2 (b, c, e, f) illustrate the lower similarity and weaker alignment of part-level text compared to object-level text [37, 53] in pretrained CLIP.

Second, current OVPS approaches struggle to achieve a comprehensive structural understanding between parts and



(a) CAT-Seg [13] (b) PartCLIPSeg [14]

Figure 3. **Challenges of Current OVPS.** (a, b) Prediction results of state-of-the-art OVSS [13] and OVPS [14] methods. Due to a lack of structural understanding, these methods often produce incorrect OVPS predictions, such as predicting a “sheep’s neck” as being larger than its “torso” or placing a “bird’s beak” and a “bird’s neck” at the tip of its “tail”.

objects. As shown in Figure 3, existing state-of-the-art models may misclassify a “leg” as a “tail” due to an incomplete understanding of part-object relationships and structural information. This happens because the visual features of the “leg” resemble those of the “tail”, leading to misrecognition. This issue is similar to the part-whole illusion [58] observed in humans when focusing only on parts without considering the whole.

We propose a novel framework, PartCATSeg, to address the critical challenges of OVPS and significantly improve model performance, as summarized in Figure 1. First, we focus on the issue of *part-level image guidance* being biased toward object-level guidance and propose a solution to mitigate this bias. Specifically, we adopt a cost aggregation strategy for explicit handling of image-text correspondence, as outlined in [11, 13]. Then, we extend this model architecture to separate and operate with cost volumes tailored to both object and part levels. This extension enables our model to disentangle and process object-level and part-level costs independently. This change provides more precise guidance and enhances image-text alignment, particularly at the part level. To further augment training signals, we integrate an unsupervised compositional loss that enforces the compositional relationship where parts collectively represent the object. This approach ensures robust part representation even with the lack of explicit part-level supervision.

Second, we leverage DINO [34, 50] features for *structural guidance* to improve the model’s ability to understand inter-part relationships and effectively separate objects from the background. Previous works, such as VLPart, highlighted DINO’s rich semantic representations and utilized them for matching novel classes to their closest base classes. Unlike their perspective, our study focuses on DINO’s capability to convey spatial structure and geometric proximity information between features. To fully exploit this rich geometric prior, we incorporate pixel-level features as guidance when conducting the cost aggregation. This pixel-level similarity between features proves particularly beneficial for

distinguishing parts.

Our approach sets a new standard baseline for OVPS, demonstrating remarkable performance gains across diverse unseen categories in multiple datasets. Extensive experiments on benchmark datasets such as Pascal-Part-116, ADE20K-Part-234, and PartImageNet show that our method significantly outperforms existing state-of-the-art methods. We achieve notable improvements in segmentation accuracy and generalization capabilities, with increases in h-IoU scores by over 10% on all datasets in the Pred-All setting. Our analyses confirm that the combination of object-aware part-level cost aggregation techniques, the compositional loss, and structural guidance from DINO contribute to these performance gains, validating the effectiveness of our proposed framework in addressing the critical challenges of OVPS.

2. Related Work

Open-Vocabulary Semantic Segmentation (OVSS). OVSS [22, 33, 43, 47, 66, 67, 72–74, 76, 77] advances beyond traditional semantic segmentation's [6, 10, 26] predefined class limitation by enabling predictions for unseen classes. OVSS methods generally rely on the shared embedding space of vision-language models (VLMs) [31, 35, 53] such as CLIP, allowing both visual and text features to be aligned in a unified semantic space. There are two main approaches. The first approach is a two-stage approach [18, 21, 23, 38, 43, 68, 69, 73], such as ZSSeg and ODISE, that first generates class-agnostic masks before VLM-based classification. The second approach is single-stage frameworks [13, 33, 47, 67, 72, 77] such as CLIPSeg and CAT-Seg that directly align pixel-level features with text features.

Dense Correspondence and Cost Aggregation. In computer vision, *cost* computation is used to evaluate *dense visual correspondence* [7, 11, 27–29, 42, 48, 49, 56, 59] by assessing the differences between matching points. From the perspective of Optimal Transport [52, 62], this cost reflects the difficulty of matching or high disparity. Minimizing disparity aligns with maximizing similarity. Hence, we interpret cost as a measure of correspondence, similar to prior notations [7, 11, 27, 28]. *Cost Aggregation* [7, 11, 12, 27, 29, 56, 59] is a key framework to reduce matching errors and enhance generalization in dense correspondence tasks. Early methods using handcrafted approaches [16, 41] transitioned to learnable kernels with 2D and 4D convolutions [27, 54]. With the advent of Transformer architectures, CATs and CATs++ [11, 12] demonstrated the effectiveness of Transformer-based cost aggregation. Recently, CAT-Seg [13] extended this concept to multi-modal settings, using cosine similarity between image and text embeddings as a matching cost volume. In this study, we apply cost aggregation to open-vocabulary part segmentation, enabling recognition of finer-grained entities.

Part Segmentation [3, 8, 8, 15, 24, 25, 30, 60] aims to recognize objects by dividing them into finer units (parts). Early works focused on contrastive and self-supervised co-part segmentation [15, 25, 30, 60], establishing foundational methods for identifying object parts without extensive labeled data. Recent developments have advancing toward open-vocabulary approaches, with notable research including VLPart [57], OV-PARTS [65], PartGLEE [36], and PartCLIPSeg [14]. VLPart aimed to generalize to novel classes using DINO correspondence [34, 50], OV-PARTS introduced object mask prompts, while PartGLEE and PartCLIPSeg combined part information with object-level context. These approaches have enhanced part segmentation performance, particularly in handling previously unseen categories. In this work, we further advance these studies by improving the image-text correspondence for both objects and parts.

3. Method

3.1. Problem Definition

Open-Vocabulary Part Segmentation (OVPS) [14, 36, 57, 65] aims to segment an image into distinct `object-specific parts`, $\mathbf{C}_{\texttt{Obj-Part}}$, (e.g., "cat's paw," "bus's headlight") without relying on predefined part categories. During training, the model uses image and ground-truth mask pairs, where the masks represent only the $\mathbf{C}_{\texttt{Obj-Part}}$ of base categories, $\mathbf{C}^{\texttt{base}}_{\texttt{Obj-Part}}$. At inference, the model segments novel categories, $\mathbf{C}^{\texttt{novel}}_{\texttt{Obj-Part}}$, beyond these base categories, leveraging learned representations to generalize without additional annotations. OVPS performance is assessed in two scenarios [14, 57, 65]: In **Zero-Shot** setup, the training and test categories do not overlap ($\mathbf{C}^{\texttt{base}}_{\texttt{Obj-Part}} \cap \mathbf{C}^{\texttt{novel}}_{\texttt{Obj-Part}} = \emptyset$). While in **Cross-Dataset** setting, the model evaluated on a different dataset from the training set without fine-tuning, introducing challenges from domain shifts and differences in part granularity.

3.2. Preliminary

Image-Text Cost Computation. As described in Section 2, *cost* refers to the correspondence between image-to-image or image-to-text features. CAT-Seg [13] extends the cost computation approach to effectively model image-text correspondence, showing promising performance in object-level open-vocabulary segmentation tasks [18, 22, 74]. The initial cost $\mathbb{C} \in \mathbb{R}^{(H\times W)\times|T|}$ between the ith image patch and the nth text token is defined as:

$$\mathbb{C}(i,n) = \frac{D^V(i)\cdot D^L(n)}{\|D^V(i)\|\|D^L(n)\|}, \tag{1}$$

where $D^V(\cdot)$ and $D^L(\cdot)$ are dense visual and language embeddings, respectively. Here, $i \in \mathbb{R}^{(H\times W)}$ is the spatial in-

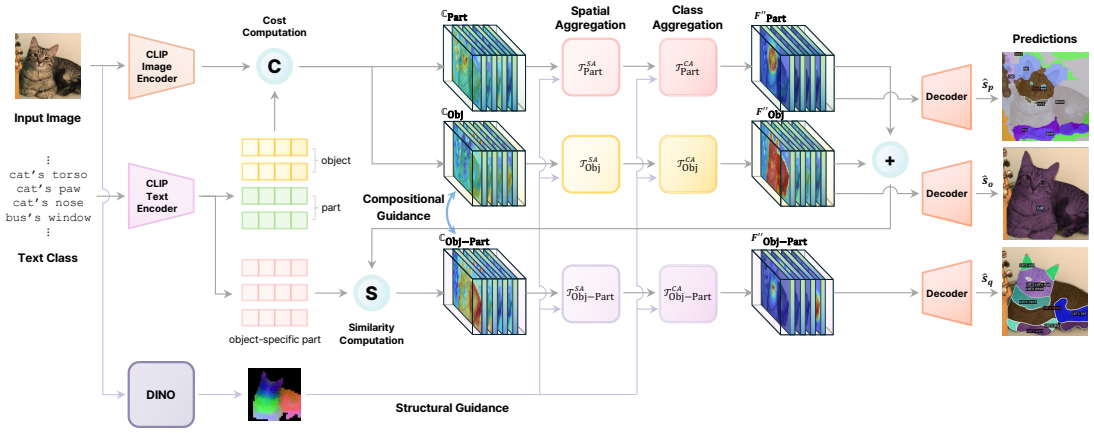


Figure 4. The Overall Architecture of PartCATSeg.

dex of the image patch and $n \in \mathbb{R}^T$ denotes the index of the text token corresponding to category classes T.

Cost Aggregation. The computed cost volume is refined through a process known as cost aggregation [7, 11, 13, 27, 28], which employs self-attention modules [61]. This process consists of two main components: the spatial aggregation transformer ($\mathcal{T}^{\text{SA}}$) and the class aggregation transformer ($\mathcal{T}^{\text{CA}}$), both of which are designed to effectively model image-text correspondence. **Spatial aggregation** utilizes the local continuity of image embeddings within the cost volume to capture spatially consistent features. Specifically, Swin Transformer [45] blocks are used to enhance semantic coherence by capturing both local and semi-global features, refining class representations, and reducing background noise. The initial cost volume feature $F \in \mathbb{R}^{(H\times W)\times|T|\times d}$ is derived from $\mathbb{C}$ using convolutional layers, represented as: $F = \texttt{Conv}(\mathbb{C})$. The spatial aggregation operation for each class is then defined as:

$$F'(:,n) = \mathcal{T}^{\text{SA}}(F(:,n)), \quad F(:,n) \in \mathbb{R}^{(H\times W)\times d}, \tag{2}$$

where $\mathcal{T}^{\text{SA}}$ is applied independently for each class in the cost volume. **Class aggregation** models inter-class relationships using a Transformer block without positional embeddings, ensuring permutation invariance. This operation accounts for the spatial context of each class and its interactions with other classes, enabling a holistic cost calculation for each class:

$$F''(i,:) = \mathcal{T}^{\text{CA}}(F'(i,:)), \quad F' \in \mathbb{R}^{(H\times W)\times|T|\times d}. \tag{3}$$

This two-stage aggregation framework refines the cost volume, enhancing prediction accuracy across diverse object categories.

3.3. Disentangled Cost Aggregation

Although CAT-Seg [13] has shown promising results in open-vocabulary semantic segmentation (OVSS), its performance is limited when it comes to fine-grained, part-level categories. This limitation arises from the predictions being biased towards object-level categories, caused by the lack of **part-level correspondence**, as shown in Figure 2.

To address this issue, we propose a method of parsing object-level and part-level class names $\mathbf{C}_{\texttt{Obj}}$ and $\mathbf{C}_{\texttt{Part}}$, computing separate cost volumes for each, similar to approaches in OVPS frameworks like PartGLEE [36] and PartCLIPSeg [14]. However, our approach differentiates itself by explicitly constructing separate cost volumes from the image-text correspondence perspective. Specifically, we compute the initial cost volumes for object-level categories and part-level categories, $\mathbb{C}_{\{\texttt{Obj|Part}\}} \in \mathbb{R}^{(H\times W)\times|\mathbf{C}_{\{\texttt{Obj|Part}\}}|}$ as follows:

$$\mathbb{C}_{\{\texttt{Obj|Part}\}}(i,x) = \frac{D^V(i)\cdot D^L(x)}{\|D^V(i)\|\|D^L(x)\|}, \; x \in \{o,p\}, \tag{4}$$

where o and p are the object-level and part-level class tokens, respectively, from the categories $\mathbf{C}_{\texttt{Obj}}$ and $\mathbf{C}_{\texttt{Part}}$.

We then obtain initial cost volume features $F_{\texttt{Obj}} \in \mathbb{R}^{(H\times W)\times|\mathbf{C}_{\texttt{Obj}}|\times d}$ and $F_{\texttt{Part}} \in \mathbb{R}^{(H\times W)\times|\mathbf{C}_{\texttt{Part}}|\times d}$ through convolutional layers, respectively. These features are further refined using spatial and class aggregation transformers to enhance the correspondence between image regions and class tokens. This allows us to compute cost volumes that are better aligned with the hierarchical concepts, providing more precise correspondence for both object-level and part-level class tokens. As shown in Figure 4, the features are processed through a sequence of spatial and class aggregation transformers as:

$$F''_{\texttt{Obj}}(i,:) = \mathcal{T}^{\text{CA}}_{\texttt{Obj}}(\mathcal{T}^{\text{SA}}_{\texttt{Obj}}(F_{\texttt{Obj}}(i,:))), \tag{5}$$

$$F''_{\texttt{Part}}(i,:) = \mathcal{T}^{\text{CA}}_{\texttt{Part}}(\mathcal{T}^{\text{SA}}_{\texttt{Part}}(F_{\texttt{Part}}(i,:))), \tag{6}$$

where i is a spatial location index. The refined cost volume embeddings $F''_{\texttt{Obj}}$ and $F''_{\texttt{Part}}$ are then fed into decoders to generate prediction masks $\hat{s}_o$ for objects and $\hat{s}_p$ for parts. These predictions are supervised using the corresponding

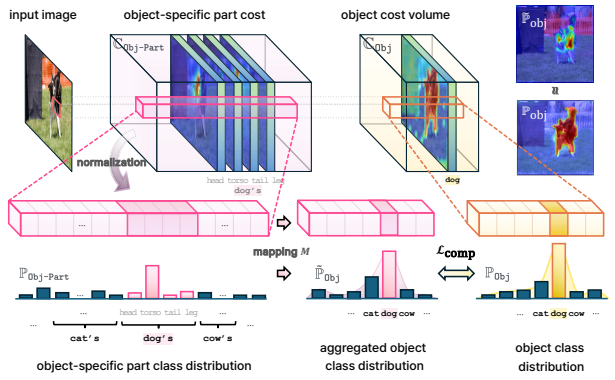


Figure 5. **Compositional Loss.** The loss function $\mathcal{L}_{\texttt{comp}}$ guides learning by ensuring that the aggregated class distribution of object parts aligns closely with the overall class distribution of the object, enhancing the consistency between part-level and object-level representations.

ground truth masks s_o, s_p with **disentanglement loss** as:

$$\mathcal{L}_{\texttt{disen}} = \lambda_{\texttt{Obj}} \sum_{o \in \mathbf{C}_{\texttt{Obj}}} \text{BCE}(s_o, \hat{s}_o) + \lambda_{\texttt{Part}} \sum_{p \in \mathbf{C}_{\texttt{Part}}} \text{BCE}(s_p, \hat{s}_p), \tag{7}$$

where BCE denotes the binary cross-entropy loss, and $\lambda_{\texttt{Obj}}$ and $\lambda_{\texttt{Part}}$ are hyperparameters balancing two losses. By leveraging this decoupled and disentangled cost aggregation strategy, our approach effectively enhances object-aware part-level image-text correspondence.

3.4. Object-aware Part Cost Aggregation

To further enhance image-text correspondence for object-specific parts, we integrate the object-level cost feature $F''_{\texttt{Obj}}$ and the part-level cost feature $F''_{\texttt{Part}}$. This integration forms an object-part combination, denoted as $\mathcal{F}_{\texttt{Obj-Part}} \in \mathbb{R}^{(H\times W)\times|\mathbf{C}_{\texttt{Obj-Part}}|\times d}$, using a simple projection as follows:

$$\mathcal{F}_{\texttt{Obj-Part}}(i) = \texttt{Linear}\left(\left[F''_{\texttt{Obj}}(i); F''_{\texttt{Part}}(i)\right]\right), \tag{8}$$

where $[\cdot;\cdot]$ denotes concatenation and $\texttt{Linear}$ is a projection layer. By combining object and part features, the model integrates object-level context with part-level details. We then align the integrated features with the object-specific part category names, $\mathbf{C}_{\texttt{Obj-Part}}$, by computing the similarity between $\mathcal{F}_{\texttt{Obj-Part}}$ and the object-specific part text embeddings $q \in \mathbf{C}_{\texttt{Obj-Part}}$, thereby obtaining $\mathbb{C}_{\texttt{Obj-Part}}$ as:

$$\mathbb{C}_{\texttt{Obj-Part}}(i,q) = \frac{\mathcal{F}_{\texttt{Obj-Part}}(i,q)\cdot D^L(q)}{\|\mathcal{F}_{\texttt{Obj-Part}}(i,q)\|\|D^L(q)\|}. \tag{9}$$

The updated object-aware part cost undergoes a convolution operation to transform it into a feature representation, $F_{\texttt{Obj-Part}}$, before refinement through spatial and class aggregation transformers.

$$F''_{\texttt{Obj-Part}}(i,:) = \mathcal{T}^{\text{CA}}_{\texttt{Obj-Part}}\left(\mathcal{T}^{\text{SA}}_{\texttt{Obj-Part}}\left(F_{\texttt{Obj-Part}}(i,:)\right)\right), \tag{10}$$

where $F_{\texttt{Obj-Part}}$ is a convoluted object-aware cost feature from $\mathbb{C}_{\texttt{Obj-Part}}$. By propagating object-level and part-level semantics between image and text, this approach improves image-text correspondence alignment, enhancing accurate boundary recognition at both object and part levels. Finally, the refined object-aware part embeddings $F_{\texttt{Obj-Part}}$ are fed into the shared decoder to generate the prediction masks $\hat{s}_q$ and supervised using the ground truth masks s_q with the following loss: $\mathcal{L}_{\texttt{Obj-Part}} = \sum_{q\in\mathbf{C}_{\texttt{Obj-Part}}} \text{BCE}\left(s_q, \hat{s}_q\right)$.

3.5. Cost as Compositional Components

Parts are *compositional components* that constitute an object. However, the ground truth annotations for parts are relatively scarce compared to those for objects, making supervision from part annotations alone insufficient.

Compositional Bias. To address this problem, we propose a compositional loss, $\mathcal{L}_{\texttt{comp}}$, that enforces inductive bias of ensuring that the parts are the compositional component of its object. This loss function enhances the model's capacity to capture spatial relationships among parts and improves the part boundary delineation, particularly for smaller components. Specifically, at each spatial location $i \in \mathbb{R}^{(H\times W)}$ in the cost volumes, we compute the normalized distributions $\mathbb{P}_{\texttt{Obj}}$ and $\mathbb{P}_{\texttt{Obj-Part}}$ over object and part classes, respectively, by applying a softmax function to $\mathbb{C}_{\texttt{Obj}}$ and $\mathbb{C}_{\texttt{Obj-Part}}$ along the class dimension as:

$$\mathbb{P}_{\texttt{Obj}}(i,o) = \frac{\exp(\mathbb{C}_{\texttt{Obj}}(i,o))}{\sum_{k\in\mathbf{C}_{\texttt{Obj}}}\exp(\mathbb{C}_{\texttt{Obj}}(i,k))}, \tag{11}$$

$$\mathbb{P}_{\texttt{Obj-Part}}(i,p) = \frac{\exp(\mathbb{C}_{\texttt{Obj-Part}}(i,p))}{\sum_{k\in\mathbf{C}_{\texttt{Obj-Part}}}\exp(\mathbb{C}_{\texttt{Obj-Part}}(i,k))}. \tag{12}$$

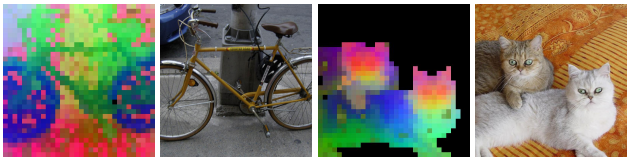
We then map the part distributions to their corresponding object classes using a predefined mapping $M : \{1,\ldots,|\mathbf{C}_{\texttt{Obj-Part}}|\} \rightarrow \{1,\ldots,|\mathbf{C}_{\texttt{Obj}}|\}$, which is deterministically defined based on $\mathbf{C}_{\texttt{Obj-Part}}$. The aggregated object class distribution $\tilde{\mathbb{P}}_{\texttt{Obj}} \in \mathbb{R}^{(H\times W)\times|\mathbf{C}_{\texttt{Obj}}|}$ is then obtained by summing the mapped part probabilities as:

$$\tilde{\mathbb{P}}_{\texttt{Obj}}(i,o) = \sum_{p\in M^{-1}(o)} \mathbb{P}_{\texttt{Obj-Part}}(i,p). \tag{13}$$

We finally compute the Jensen-Shannon [19, 39] divergence between the object's distribution and the aggregated object distribution at each spatial location:

$$\mathcal{L}_{\texttt{comp}} = \frac{1}{2}\left(D_{\text{KL}}(\mathbb{P}_{\texttt{Obj}}\|\tilde{\mathbb{P}}_{\texttt{Obj}}) + D_{\text{KL}}(\tilde{\mathbb{P}}_{\texttt{Obj}}\|\mathbb{P}_{\texttt{Obj}})\right), \tag{14}$$

By enforcing the similarity between the object's distribution and the aggregated object distribution, we inject the inductive bias that parts collectively compose the object. This encourages the model to recognize the spatial relationships among parts and ensures that smaller parts are appropriately delineated.



(a) w/o mask (b) w/ mask

Figure 6. **Visualization of Frozen DINO PCA results.** (a) Without mask: The PCA visualization shows that DINO can separate object features from the background. (b) With mask: highlights consistent feature values across structurally similar parts of the objects, indicating strong part correspondences. This showcases DINO's ability to capture detailed object representations.

Ultimately, our training objective aims to minimize the following loss function:

$$\mathcal{L} = \mathcal{L}_{\texttt{Obj-Part}} + \mathcal{L}_{\texttt{disen}} + \lambda_{\texttt{comp}}\mathcal{L}_{\texttt{comp}}, \quad (15)$$

where $\lambda_{\texttt{comp}}$ is a hyperparameter for balancing the loss.

3.6. Cost with Structural Guidance

We utilize DINO's [5, 50] self-supervised learning capabilities, which are known to exhibit strong semantic and geometric priors. DINO features have previously demonstrated their effectiveness in VLPart [57] by aiding in the correspondence computation between base and novel classes. While VLPart focused on image-to-image correspondences, we extend this approach by forming an image-text cost volume, enabling the use of DINO features for image-text correspondence directly related to specific categories.

We employ DINO features to provide structural guidance during spatial aggregation. As shown in Figure 6 (a), the PCA visualization illustrates that DINO features effectively distinguish between objects and background components. Additionally, in Figure 6 (b), after background masking and PCA application, we observed detailed intra-object information, revealing continuous PCA results within objects. This indicates that DINO features effectively capture structural information useful for predicting part relationships. The visualization further shows consistent representations for structurally similar parts (e.g., "heads" in red, "torso" in blue, and "legs" in green for two "cats"), demonstrating the alignment capability of DINO features.

Structural Guidance. To maximize the use of these structural features, we introduce structural guidance during the spatial aggregation process by incorporating it into the $Query$ and Key matrices along with the cost volume as:

$$F'(:,n) = \mathcal{T}^{\texttt{SA}}(F(:,n), F_{\texttt{dino}}(:,n)) \in \mathbb{R}^{(H\times W)\times(d+d_{\texttt{dino}})}. \quad (16)$$

The effectiveness of our approach is comprehensively evaluated in the ablation study (see Section 4.3). The study demonstrates the impact of incorporating structural guidance into the cost volume, leading to improved part and object image-text correspondences.

4. Experiments

4.1. Experimental Setup

Datasets. We perform our evaluation on three OVPS benchmarks: Pascal-Part-116 [8, 65], ADE20K-Part-234 [65, 75], and PartImageNet [24]. Pascal-Part-116 is an adapted version of PascalPart [8] of OVPS, containing 8k training images and 850 test images, with 116 part classes across 17 object categories. ADE20K-Part-234, which is a more challenging and fine-grained dataset, includes 7k training images and 1k validation images with 234 part classes in 44 object categories. PartImageNet provides 16k training images and 2.9k validation images across 158 object classes from ImageNet [17], with 40 representative classes used for cross-dataset evaluation in this study. Additionally, we use PartImageNet (OOD), a split originally designed for few-shot learning, consisting of 109 training, 19 validation, and 30 test classes. This split is included for cross-dataset evaluation. Additional details of the datasets are provided in the supplementary materials.

Evaluation Metrics. To comprehensively evaluate OVPS performance, we use two protocols: (1) **Pred-All** introduced by PartCLIPSeg [14], where the model performs segmentation without access to object masks or class labels, relying entirely on its own predictions, and (2) **Oracle-Obj**, where ground truth object-level masks and class labels are provided as in setting of OV-PARTS [65]. This setup assumes the scenario of part segmentation with the reliance on an object-level segmentation results. We measure segmentation performance using mIoU and assess generalization via the harmonic mean (h-IoU) of seen and unseen categories.

Implementation Details. We adopted the training approach of CAT-Seg [13] and fine-tuned the Query and Value heads of CLIP's encoders. Details on the information on the experimental setup, hyperparameters, and training can be found in the supplementary materials.

4.2. Main Results

Zero-Shot Part Segmentation. We compare PartCATSeg with previous methods on three popular OVPS benchmarks, demonstrating its significant performance and strong generalization capabilities. As shown in Table 1, PartCATSeg consistently outperforms existing methods on Pascal-Part-116, with significant improvements in h-IoU: a 15.10% increase in Pred-All and 11.62% increase in Oracle-Obj over the second-best method, PartCLIPSeg [14]. On the more challenging ADE20K-Part-234 dataset, as demonstrated in Table 2, PartCATSeg again outperforms previous SoTA methods, achieving a remarkable 12.81% improvement in Pred-All and 8.13% in Oracle-Obj. On PartImageNet, as in Table 3, PartCATSeg sets new performance benchmarks, breaking through the h-IoU thresholds with 55.12% in Pred-All and 72.66% in Oracle-Obj. These re-

Method	Pred-All			Oracle-Obj		
	Seen	Unseen	h-IoU	Seen	Unseen	h-IoU
ZSSeg+ [69]	38.05	3.38	6.20	54.43	19.04	28.21
VLPart [57]	35.21	9.04	14.39	42.61	18.70	25.99
CLIPSeg [47, 65]	27.79	13.27	17.96	48.91	27.54	35.24
CAT-Seg [13, 65]	36.80	23.39	28.60	43.81	27.66	33.91
PartGLEE [36]	-	-	-	57.43	27.41	37.11
PartCLIPSeg [14]	43.91	23.56	30.67	50.02	31.67	38.79
PartCATSeg (Ours)	**52.62**	**40.51**	**45.77** (+15.10)	**57.49**	**44.88**	**50.41** (+11.62)

[1] The best score is **bold** and the second-best score is underlined.

Table 1. Comparison of zero-shot performance with state-of-the-art methods on Pascal-Part-116.

Method	Pred-All			Oracle-Obj		
	Seen	Unseen	h-IoU	Seen	Unseen	h-IoU
ZSSeg+ [69]	32.20	0.89	1.74	43.19	27.84	33.85
CLIPSeg [47, 65]	3.14	0.55	0.93	38.15	30.92	34.15
CAT-Seg [13, 65]	7.02	2.36	3.53	33.80	25.93	29.34
PartGLEE [36]	-	-	-	51.29	35.33	41.83
PartCLIPSeg [14]	14.15	9.52	11.38	38.37	38.82	38.60
PartCATSeg (Ours)	**38.87**	**17.56**	**24.19** (+12.81)	**53.13**	**47.16**	**49.96** (+8.13)

Table 2. Comparison of zero-shot performance with state-of-the-art methods on ADE20K-Part-234.

Method	Pred-All			Oracle-Obj		
	Seen	Unseen	h-IoU	Seen	Unseen	h-IoU
CLIPSeg [47, 65]	32.39	12.27	17.80	53.91	37.17	44.00
CAT-Seg [13, 65]	33.58	23.04	27.33	47.34	35.14	40.33
PartCLIPSeg [14]	38.82	19.47	25.94	56.26	51.65	53.85
PartCATSeg (Ours)	**57.33**	**53.07**	**55.12** (+27.79)	**73.83**	**71.52**	**72.66** (+18.81)

Table 3. Comparison of zero-shot performance with state-of-the-art methods on PartImageNet.

sults highlight PartCATSeg's advanced capabilities, particularly in performance on unseen categories, positioning it as a leading solution in OVPS with exceptional accuracy in both seen and unseen classes. By addressing the challenges in OVPS, our method achieves a significant advancement in OVPS, highlighting PartCATSeg's robustness and superior generalization abilities across diverse datasets.

Cross-Dataset Part Segmentation. Our experiments evaluate the performance of our method in two cross-dataset settings: (1) PartImageNet (OOD) and (2) PartImageNet → Pascal-Part-116. PartImageNet (OOD) includes classes with complex, irregular part shapes that differ from the training data, for evaluating adaptability to novel configurations within a familiar structure. As shown in Table 4, PartCATSeg outperforms previous methods in both Pred-All and Oracle-Obj settings, demonstrating its robustness in handling unseen categories in an OOD context. On PartImageNet → Pascal-Part-116, we evaluate the model's ability to transfer to a different dataset, Pascal-Part-116 [8, 65]. Trained on PartImageNet, our model is evaluated on Pascal-Part-116. Herein, PartCATSeg again achieves the highest performance across both Pred-All and Oracle-Obj settings. These results confirm PartCATSeg's effectiveness in diverse cross-dataset scenarios, highlighting its versatility and robustness in part segmentation tasks across various dataset configurations.

Qualitative Evaluation. We present qualitative results on the

Method	PartImageNet (OOD)		PartImageNet → Pascal-Part-116	
	Pred-All	Oracle-Obj	Pred-All	Oracle-Obj
	Unseen	Unseen	Unseen	Unseen
CLIPSeg [47, 65]	6.69	55.86	11.72	14.87
CAT-Seg [13, 65]	19.83	39.12	11.55	12.50
PartCLIPSeg [14]	10.33	59.16	14.74	19.86
PartCATSeg (Ours)	**40.17** (+20.34)	**66.15** (+6.99)	**18.21** (+3.47)	**22.88** (+3.02)

Table 4. Cross-dataset performance with PartImageNet (OOD) and PartImageNet to Pascal-Part-116.

Compositional Loss	Pred-All			Oracle-Obj		
	Seen	Unseen	h-IoU	Seen	Unseen	h-IoU
Cost Agg	44.05	25.06	31.94	56.40	28.26	37.66
+ DINO	54.48	35.89	43.28	60.56	40.51	48.55
+ DINO + $\mathcal{L}_{\texttt{comp-L1}}$	**54.70**	35.92	43.36	**61.29**	41.62	49.57
+ DINO + $\mathcal{L}_{\texttt{comp-SM}}$	52.62	**40.51**	**45.77**	57.49	**44.88**	**50.41**

Table 5. Impact of Compositional Loss on Pascal-Part-116

Structural Guidance	Pred-All			Oracle-Obj		
	Seen	Unseen	h-IoU	Seen	Unseen	h-IoU
w/o Structural Guidance	42.29	27.94	33.65	46.44	31.59	37.60
DINO ($\mathcal{T}^{\texttt{SA}}_{\texttt{Obj}}$)	50.24	31.35	38.61	55.03	34.29	42.25
DINO ($\mathcal{T}^{\texttt{SA}}_{\texttt{Part}}$)	**56.28**	36.67	44.41	**62.73**	43.47	**51.35**
DINO ($\mathcal{T}^{\texttt{SA}}_{\texttt{Obj}}$, $\mathcal{T}^{\texttt{SA}}_{\texttt{Part}}$)	52.62	**40.51**	**45.77**	57.49	**44.88**	50.41

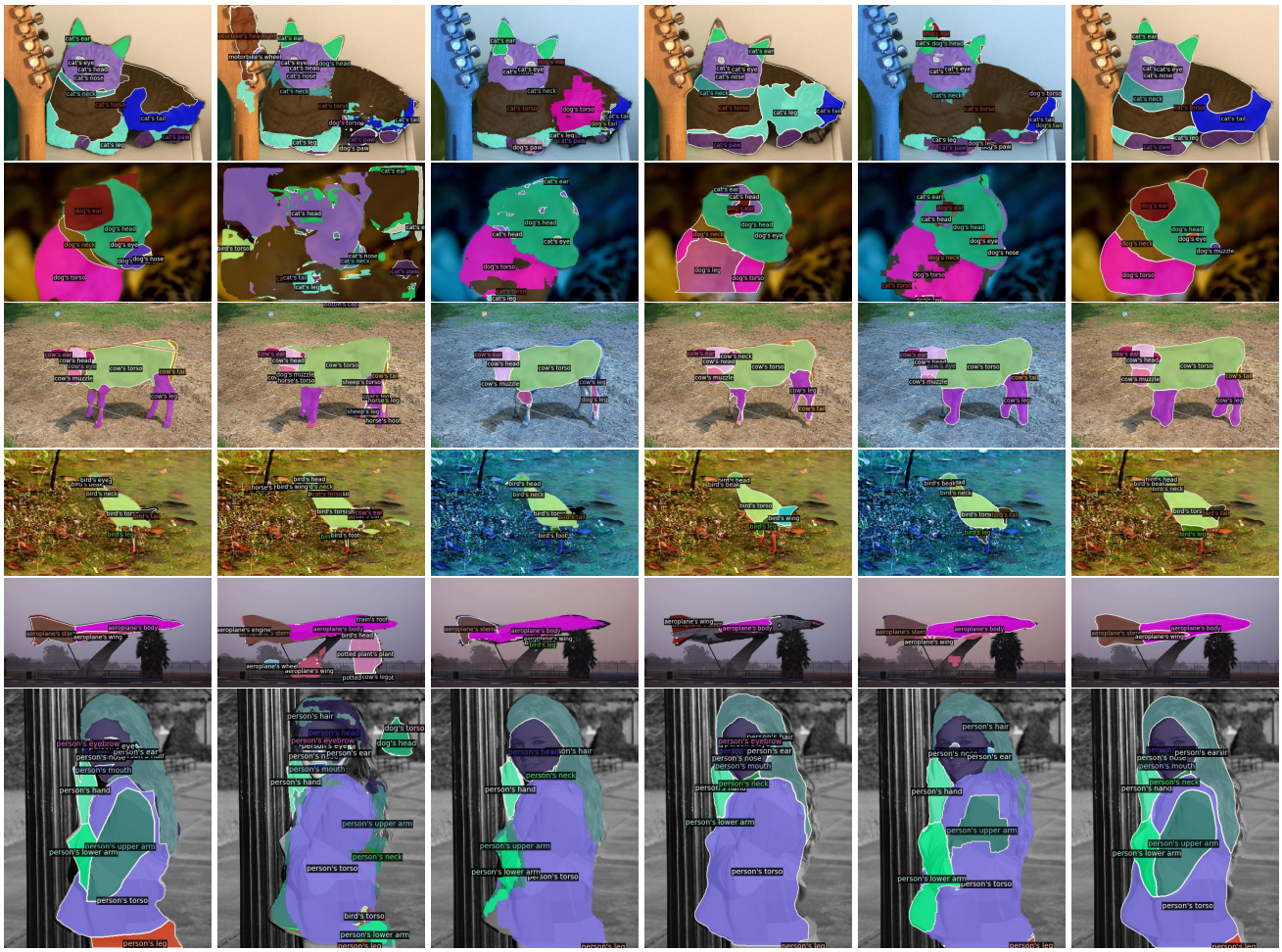
[1] All configurations use the compositional loss $\mathcal{L}_{\texttt{comp}}$.
[2] Configurations utilizing DINO also include object-specific part-level guidance.

Table 6. Impact of Structural Guidance on Pascal-Part-116.

Pascal-Part-116 dataset in the Pred-All setting. Figure 7 compares our method with other baselines, demonstrating significant improvements by addressing several challenges in previous OVPS methods. For instance, in **rows 1 and 2**, compared to the other baselines showing unnatural segmentations, our model successfully segments the cat's leg and the dog's neck, preserving the object's structural integrity with the structural guidance. In **row 6**, PartCATSeg correctly segments the person's upper and lower arm as distinct parts. Additionally, our method almost completely segments the parts in the object without leaving regions unsegmented and effectively captures smaller parts that other models often miss. For example, in **rows 2 – 4**, finer details like the dog's ear, the cow's tail, and the bird's beak are accurately segmented in our method. In **rows 5 and 6**, our method segments almost all the parts in the object without sacrificing accuracy compared to the other baselines. This improvement is largely due to the compositional loss, which encourages the competitive assignment of parts that collectively constitute the whole object. Moreover, our method resolves confusion between similar object categories. In **rows 1 and 2**, other models misclassify parts of the cat and dog, sometimes confusing the cat for a dog and vice versa. Our object-aware part-level guidance helps the model correctly segment the parts of the correct object.

4.3. Ablation Study

Impact of Compositional Loss. To assess the effectiveness of the proposed compositional loss $\mathcal{L}_{\texttt{comp}}$, we conduct an ablation study on its impact on Pascal-Part-116 in Table 5. We compare the effects of using compositional loss with a softmax-based normal-



(a) Ground-truth (b) VLPart [57] (c) CLIPSeg [53, 65] (d) CAT-Seg [13, 65] (e) PartCLIPSeg [14] (f) PartCATSeg

Figure 7. Qualitative evaluation of zero-shot part segmentation on Pascal-Part-116 in the **Pred-All** configuration. Note that annotations for unseen categories (e.g., bird, cow, dog) are excluded from the training set.

ization ($\mathcal{L}_{\texttt{comp-SM}}$), which encourages each spatial location to be predominantly assigned to a single part class, versus L1 normalization ($\mathcal{L}_{\texttt{comp-L1}}$) before combining part costs. We observe that adding $\mathcal{L}\texttt{comp}$ gains about 2% h-IoU increase in the Pred-All setting compared to the model without using $\mathcal{L}\texttt{comp}$. Furthermore, the model using $\mathcal{L}_{\texttt{comp-SM}}$ outperforms the one using $\mathcal{L}_{\texttt{comp-L1}}$. These results confirm that $\mathcal{L}_{\texttt{comp}}$ effectively improves performance by leveraging the inductive bias that parts collectively compose the object. Further qualitative results of ablation on $\mathcal{L}_{\texttt{comp}}$ are provided in the supplementary materials.

Impact of Structural Guidance from DINO. To investigate the effect of structural guidance from DINO, we conduct an ablation study by selectively applying it in our cost aggregation framework. In PartCATSeg, structural guidance from DINO is incorporated into three levels of spatial aggregation transformer modules: objects ($\mathcal{T}^{\texttt{SA}}_{\texttt{Obj}}$), parts ($\mathcal{T}^{\texttt{SA}}_{\texttt{Part}}$), and object-specific parts ($\mathcal{T}^{\texttt{SA}}_{\texttt{Obj-Part}}$). We measure performance changes on Pascal-Part-116 when structural guidance is applied only at the object level, only at the part level, and at both levels. As shown in Table 6, applying structural guidance at the part level yields more h-IoU increases in both Pred-All and Oracle-Obj settings than at the object level. When structural guidance is applied at both levels, we observe further improvements, especially in unseen classes. This suggests that, within the cost aggregation framework, structural guidance from DINO is more effective when leveraging detailed structural information within individual objects, rather than relying solely on DINO's background-object separation ability.

5. Conclusion

In this paper, we present a novel approach to open-vocabulary part segmentation, PartCATSeg, which leverages cost aggregation to address the inherent challenges of OVPS. We make strategic modifications to cost aggregation to enhance object-aware part-level guidance, incorporate a compositional loss to compensate for limited part-level guidance from the base set, and utilize DINO features for robust structural guidance. These combined efforts enable PartCATSeg to significantly outperform existing state-of-the-art methods. We believe that our approach can serve as a powerful new baseline for practitioners and researchers, fostering greater interest and advancement within the community.

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